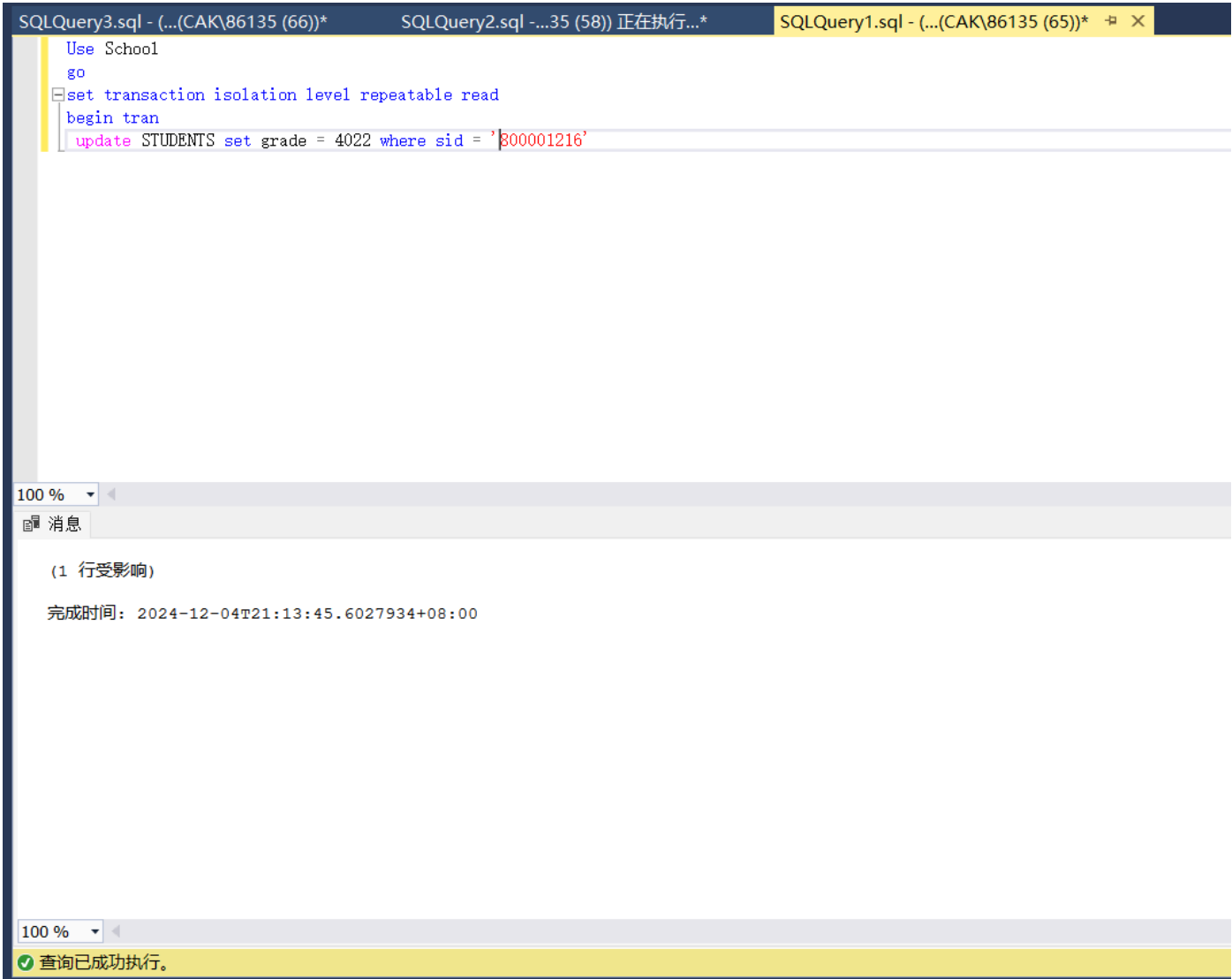


数据库实验报告 第15周

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1 创建查询1，设置隔离等级为repeatable read 不提交事务，然后运行，接着创建查询2，设置隔离等级为repeatable read，查询STUDENTS表中的记录，结果运行一直处于执行中：

查询1：



查询2：

SQLQuery3.sql - (...CAK\86135 (66))* SQLQuery2.sql - ...35 (58)) 正在执行...* SQLQuery1.sql - (...CAK\86135 (65))*

```
Use School
go
set transaction isolation level repeatable read
begin tran
select * from STUDENTS
commit tran
```

100 %

结果 消息

正在执行查询...

新建查询执行exec sp_who, 发现进程58 (查询2) 已经被进程65 (查询1) 阻塞

SQLQuery3.sql - (...CAK\86135 (66))* SQLQuery2.sql - ...35 (58)) 正在执行...* SQLQuery1.sql - (...CAK\86135 (65))*

```
exec sp_who
```

100 %

结果 消息

	spid	ecid	status	loginame	hostname	blk	dbname	cmd	request_id
41	41	0	background	sa		0	master	CHECKPOINT	0
42	42	0	background	sa		0	master	BRKR TASK	0
43	43	0	background	sa		0	master	BRKR TASK	0
44	44	0	sleeping	sa		0	master	TASK MANAGER	0
45	45	0	sleeping	sa		0	master	TASK MANAGER	0
46	46	0	sleeping	sa		0	master	TASK MANAGER	0
47	47	0	sleeping	sa		0	master	TASK MANAGER	0
48	48	0	sleeping	sa		0	master	TASK MANAGER	0
49	49	0	sleeping	sa		0	master	TASK MANAGER	0
50	50	0	sleeping	sa		0	master	TASK MANAGER	0
51	51	0	sleeping	NT SE...	CAK	0	master	AWAITING COMMAND	0
52	52	0	sleeping	sa		0	master	TASK MANAGER	0
53	53	0	sleeping	sa		0	master	TASK MANAGER	0
54	54	0	sleeping	sa		0	master	TASK MANAGER	0
55	55	0	sleeping	sa		0	master	TASK MANAGER	0
56	56	0	sleeping	sa		0	master	TASK MANAGER	0
57	57	0	sleeping	sa		0	master	TASK MANAGER	0
58	58	0	suspended	CAK\8...	CAK	65	School	SELECT	0
59	59	0	sleeping	sa		0	master	TASK MANAGER	0
60	60	0	sleeping	sa		0	master	TASK MANAGER	0
61	61	0	sleeping	sa		0	master	TASK MANAGER	0

在查询2中通过lock_timeout设置锁定超时时间间隔。在超过0.5s后，锁管理器自动解除锁的争夺。

SQLQuery4.sql - (...(CAK\86135 (65))* SQLQuery3.sql - (...(CAK\86135 (66))* SQLQuery2.sql - (...(CAK\86135 (58))*

```
Use School
go
set transaction isolation level repeatable read
set lock_timeout 500
begin tran
select * from STUDENTS
commit tran
```

100 %

结果 消息

消息 1222, 级别 16, 状态 51, 第 6 行
已超过了锁请求超时时段。

完成时间: 2024-12-04T21:29:07.5217102+08:00

再次执行exec sp_who, 65进程已经不再阻塞58进程，解锁成功：

SQLQuery4.sql - (...(CAK\86135 (65))* SQLQuery3.sql - (...(CAK\86135 (66))* SQLQuery2.sql - (...(CAK\86135 (58))*

```
exec sp_who
```

100 %

结果 消息

	spid	ecid	status	loginame	hostname	blk	dbname	cmd	request_id
46	46	0	sleeping	sa		0	master	TASK MANAGER	0
47	47	0	sleeping	sa		0	master	TASK MANAGER	0
48	48	0	sleeping	sa		0	master	TASK MANAGER	0
49	49	0	sleeping	sa		0	master	TASK MANAGER	0
50	50	0	sleeping	sa		0	master	TASK MANAGER	0
51	51	0	sleeping	NT SE...	CAK	0	master	AWAITING COMMAND	0
52	52	0	sleeping	sa		0	master	TASK MANAGER	0
53	53	0	sleeping	sa		0	master	TASK MANAGER	0
54	54	0	sleeping	sa		0	master	TASK MANAGER	0
55	55	0	sleeping	sa		0	master	TASK MANAGER	0
56	56	0	sleeping	sa		0	master	TASK MANAGER	0
57	57	0	sleeping	sa		0	master	TASK MANAGER	0
58	58	0	sleeping	CAK\8...	CAK	0	School	AWAITING COMMAND	0
59	59	0	sleeping	sa		0	master	TASK MANAGER	0
60	60	0	sleeping	sa		0	master	TASK MANAGER	0

2 两个连接同时执行如下查询，结果如下：

```
Use School
go
set transaction isolation level repeatable read
begin tran
select * from STUDENTS where sid = '800001216'
waitfor delay '00:00:20'
update STUDENTS set grade = 1111 where sid = '800001216'
commit tran
select * from STUDENTS where sid = '800001216'
```

100 %

结果 消息

	sid	sname	email	grade
1	800001216	gfxrgs	hhce4@qhldj.gov	2022

	sid	sname	email	grade
1	800001216	gfxrgs	hhce4@qhldj.gov	1111

```
SQLQuery6.sql - (...CAK\86135 (65))*  X SQLQuery5.sql - (...CAK\86135 (58))*
Use School
go
set transaction isolation level repeatable read
begin tran
select * from STUDENTS where sid = '800001216'
waitfor delay '00:00:20'
update STUDENTS set grade = 1111 where sid = '800001216'
commit tran
select * from STUDENTS where sid = '800001216'
```

100 %

结果 消息

(1 行受影响)
消息 1205, 级别 13, 状态 51, 第 7 行
事务(进程 ID 65)与另一个进程被死锁在 锁 资源上, 并且已被选作死锁牺牲品。请重新运行该事务。

完成时间: 2024-12-04T21:38:54.9600932+08:00

其中一个连接查询成功，另一个连接由于死锁停止查询，发出错误代码1205，表示该事务被牺牲。由于两个进程都释放不了共享锁，因此导致死锁。

3 避免死锁：

- (1) 按照顺序访问资源。当事务按照一定的顺序访问资源时，死锁就会得以避免。
- (2) 事务开始时尝试一次性获取所有锁，若无法获取，则不执行事务，来避免死锁。
- (3) 减小事务持有锁的时间。通过快速释放锁来降低死锁的风险。
- (4) 使用低隔离级别也可以可以减少锁的争用，从而降低死锁的风险。

死锁的处理（构想）：

- (1) 主动回滚部分代价比较小的事务来解除死锁。
- (2) 设置超时机制来使死锁事务回滚来解决死锁
- (3) 可以设置动态调整隔离等级来在死锁时释放锁来摆脱死锁
- (4) 合理优化查询语句
- (5) 组织对资源的有序访问